KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY (KIIT)

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CAPSTONE PROJECT REPORT

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ORDER-TO-CASH (O2C) PROCESS FLOW IN SAP

Fig. 1 : End-to-End Order-to-Cash (O2C) Process Flow in SAP

1. INTRODUCTION

SAP is the world's leading ERP platform, used by over 400,000 organizations to manage integrated business operations across finance, logistics, and sales. The Sales & Distribution (SD) module governs the complete commercial lifecycle — from customer inquiry through order fulfillment, delivery, invoicing, and payment. The Order-to-Cash (O2C) process represents this end-to-end sales cycle and is one of the most strategically critical workflows in any sales-driven organization, directly impacting revenue realization and customer experience.

2. PROBLEM STATEMENT

Organizations relying on manual, disconnected systems for sales processing face significant operational challenges:

- Data entered separately across sales, warehouse, and finance teams causes frequent mismatches and revenue leakage.
- Lack of real-time inventory visibility leads to overselling, delayed deliveries, and customer dissatisfaction.
- Manual invoice preparation is error-prone and lengthens Days Sales Outstanding (DSO).
- No automated credit control exposes the business to bad debt risk from high-risk customers.
- Disconnected systems make audit trails and compliance reporting unreliable.

This project implements the complete O2C cycle in SAP SD to address these issues through process standardization, real-time integration, and automation.

3. SOLUTION / IMPLEMENTATION FEATURES

The solution configures and executes a full end-to-end O2C process in SAP SD for a fictitious company, VKJ Trading Co., covering the following steps:

3.1 Organizational Structure (Company Blueprint)

Org. Element	Configured Value
Company Code	VKJ1 - VKJ Trading Co. Pvt. Ltd.
Sales Organization	1000 - VKJ Sales India
Distribution Channel	10 - Direct Sales
Division	00 - Cross-Division
Plant	1000 - VKJ Main Plant, Bhubaneswar
Shipping Point	SHP1 - Central Dispatch Hub
Storage Location	0001 - Main Warehouse

3.2 Step-by-Step O2C Process

Step 1 — Customer Master Creation (XD01): The Customer Master stores address, payment terms (reconciliation account in FI), and sales area data. It is mandatory before any sales document can be created.

Fig. 2 : Customer Master (XD01): Accounting Information screen showing Reconciliation Account 150000

Step 2 — Material Master (MM01): Captures product data across SD, MM, and MRP views. The sales org. view defines the delivering plant and item category group; the MRP view governs replenishment.

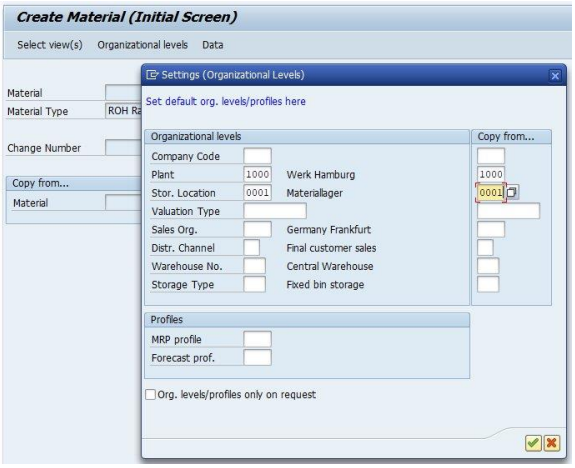

Fig. 3 : Material Master Creation (MM01): Organizational levels including plant, storage location, and sales area configuration

Step 3 — Inquiry & Quotation (VA11 / VA21): A customer Inquiry captures product interest; the system generates a formal Quotation with agreed pricing and validity, which is referenced when creating the sales order.

Step 4 — Sales Order Creation (VA01): The central O2C document. Created with document type OR, it captures sold-to/ship-to party, line items, pricing conditions, payment terms, and incoterms.

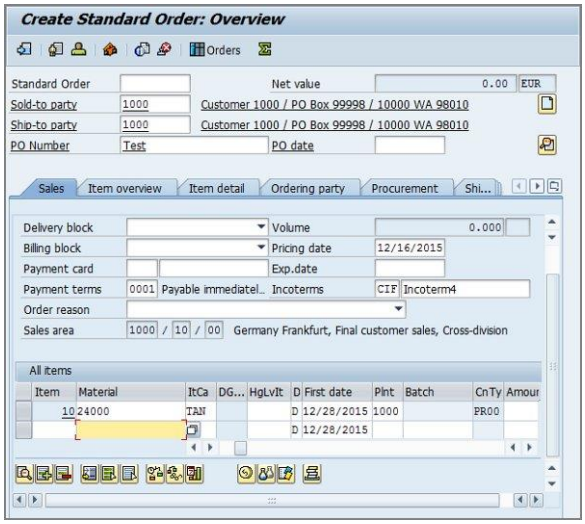

Fig. 4 : Sales Order Creation (VA01): Order Type OR, Sold-to Party 1000, CIF incoterms and pricing date

Step 5 — Availability Check / ATP (CO09): SAP automatically runs Available-to-Promise logic at order save, checking stock against confirmed demands (purchase orders, reservations, deliveries) to commit a delivery date.

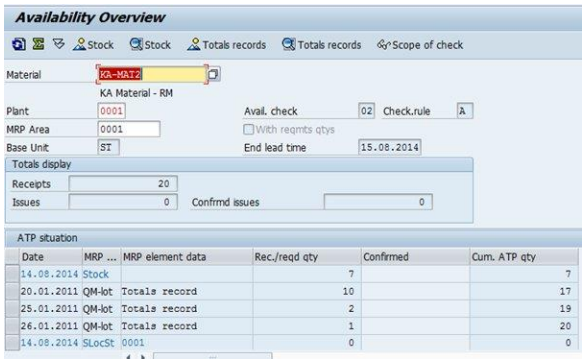

Fig. 5 : Availability Overview (CO09): Material KA-MAI2 with cumulative ATP quantities and stock receipts

Step 6 — Outbound Delivery (VL01N / VL02N): Delivery document triggers warehouse picking and packing. Post Goods Issue (PGI) reduces inventory in MM, automatically posts Cost of Goods Sold (COGS) in FI without manual entry, and unlocks the billing due list for invoice creation.

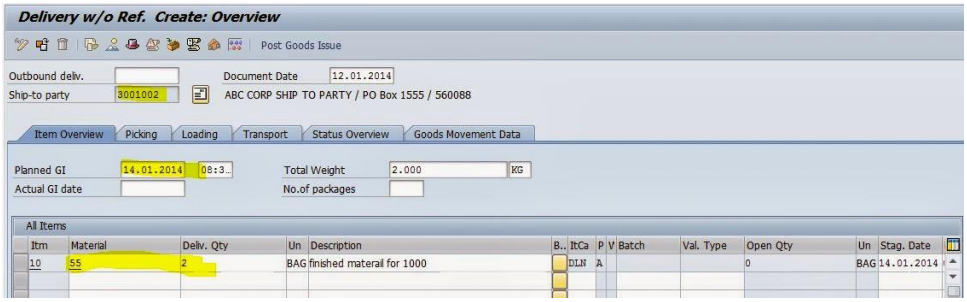

Fig. 6 : Outbound Delivery (VL01N): Create Overview showing Ship-To Party, Planned GI Date, delivery item with quantity, and Post Goods Issue trigger

Step 7 — Billing & Payment (VF01 / F-28): VF01 generates the customer invoice, posting Accounts Receivable (Dr) and Revenue (Cr) simultaneously in FI. The output determination framework auto-generates the invoice PDF. Payment is received via F-28, clearing the open AR item and completing the O2C cycle.

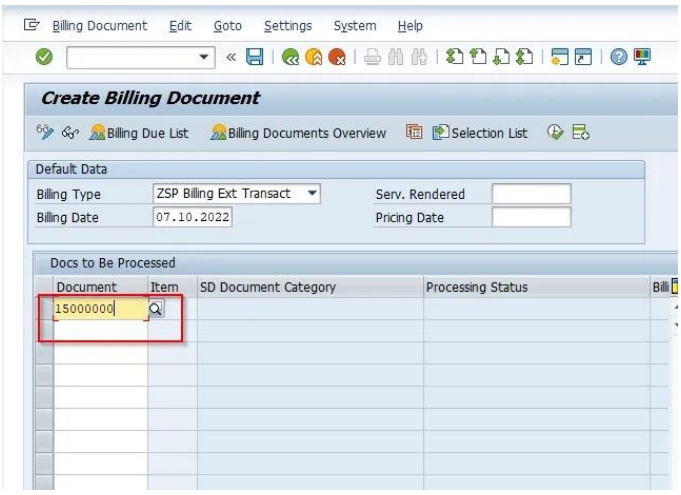

Fig. 7 : Create Billing Document (VF01): Billing Type ZSP with delivery document queued for invoice processing

4. TECH STACK

Module	Full Name	Role in O2C
SAP SD	Sales & Distribution	Core module - Inquiry, Quotation, Sales Order, Delivery, Billing
SAP MM	Materials Management	Material Master, inventory management, ATP stock data, Goods Issue
SAP FI	Financial Accounting	Customer reconciliation, A/R posting, revenue recognition, payment clearing
SAP WM	Warehouse Management	Transfer orders, picking confirmation, storage bin management

Key Transaction Codes

T-Code	Description	Additional T-Codes / Process Step
XD01	Create Customer Master	VA01 - Create Sales Order
MM01	Create Material Master	VL01N - Create Outbound Delivery
VA11	Create Inquiry	VL02N - Post Goods Issue
VA21	Create Quotation	VF01 - Create Billing Document
CO09	Availability Overview (ATP)	F-28 - Incoming Payment

5. UNIQUE POINTS / KEY FEATURES

Business Value	Technical Differentiation
1. Real-time stock visibility prevents overselling and stockouts. 2. Automated credit limit check reduces bad debt exposure significantly. 3. Single source of truth eliminates data mismatches across departments. 4. Faster billing cycle shortens DSO and improves cash flow. 5. Full document trail ensures audit readiness and regulatory compliance.	1. ATP logic integrates SD + MM in real time for reliable delivery commitment. 2. PGI triggers automatic COGS recognition — no manual FI entry needed. 3. Output determination framework auto-generates and emails invoice PDFs. 4. Pricing condition technique supports complex discounts and surcharge rules. 5. BAPI/IDOC interfaces enable seamless integration with external e-commerce.

6. FUTURE IMPROVEMENTS

- SAP S/4HANA Migration - Leverage Fiori UI and in-memory HANA for real-time order analytics and sub-second processing.
- EDI / IDOC Integration - Automated order and invoice exchange with customer systems using EDIFACT/X12 message standards.
- SAP Analytics Cloud - Live O2C dashboards tracking order backlog, delivery performance, and DSO KPIs.
- ML-Based ATP Forecasting - Predictive availability using SAP IBP demand forecasting integrated with the ATP engine.
- Collections Automation - Dunning letters and collection workflows via SAP Collections Management to minimize overdue receivables.

7. CONCLUSION

This capstone project presents a detailed, practical demonstration of the Order-to-Cash (O2C) cycle in SAP SD using VKJ Trading Co. as a conceptual case scenario. It encompasses the complete process—from enterprise structure setup and master data creation (XD01, MM01) to inquiry and quotation (VA11/VA21), sales order processing (VA01), availability check (CO09), outbound delivery with Post Goods Issue (VL01N), billing (VF01), and final payment clearing (F-28). Each step illustrates how SAP integrates sales, logistics, and financial processes to eliminate data silos and streamline revenue realization.

The interaction between SAP SD, MM, FI, and WM modules highlights the system's capability to transform traditionally fragmented manual processes into a unified, real-time, and audit-ready framework. Overall, this project reflects a strong practical understanding of document flow, cross-module integration, and the business value of real-time ATP and automated COGS posting—core competencies developed through the SAP Data/Analytics Engineer program at KIIT.